Mini-Split A/C Policy

This policy defines the proper installation of mini-split air conditioning (A/C) units to ensure safety and aesthetic harmony.

Introduction

Mini-Split A/C's can be an alternative to Window A/C units, but the installation of which may involve significant exterior, interior, and electrical modifications to your unit. Please review this policy before submitting an Mini-split A/C installation request. Please note that there is a separate "Window A/C Policy (2019)" for Window A/C installations.

Terminology

Mini-split A/C's comprise of two or more main components:

- *Outdoor unit* (also referred to as the Condenser/Compressor/Heat Pump) - Henceforth referred to as the *outdoor unit*
- *Indoor unit* (also referred to as the Air Handling Unit/Evaporator) – Henceforth referred to as the *indoor unit*

Depending on the capabilities of the (a) outdoor unit, it can support one or more (b) indoor units.

Mini-split A/C's are also known as ductless A/C systems, where instead of channeling cool air through relatively large ducts (which are not present in our Eichler units), relatively smaller tubing is used to carry liquid refrigerant between the outdoor and indoor units.

An electrical connection is required to the outdoor unit, which then powers all its connected indoor units. The refrigerant pipes and electrical cables between the outdoor and indoor unit is known as the line set.

Henceforth, the term *conduit* will be used to refer to exterior-visible parts of line sets, electrical pipe, drain pipes, or related covers to cover these components.

Locations allowed

The outdoor unit is allowed in these locations:

2BR: Backyard of unit

3BR: Backyard of unit
Atrium of unit

4BR: Backyard of unit
Front yard of unit

The outdoor unit is normally mounted to the ground or floor. For structural reasons, it shall not be mounted to the wall, ceiling, nor roof of the unit.

For outdoor units located in the backyard or front yard, the top of the outdoor unit must not be above the fence line (nominally 7 feet above ground) of the unit. Any associated electrical control panel box (for connecting or disconnecting to the outdoor unit) may be mounted on the wall of the unit, but the top of the panel must also be below the fence line of the unit.

Only wall-mounted (not ceiling-mounted) indoor units are permitted. They can be positioned on a wall closer to the ceiling and close to where the line sets can be routed to. There should be sufficient clearance between the top of the indoor unit and the ceiling.

Electrical connections and Line sets are allowed indoors and outdoors, including above the fence line of a unit, or along the exterior wall in the car port area, or along the side wall of the unit, subject to aesthetic and structural considerations (detailed below), but not over or through the roof.

For example, if it is desired to install one or more indoor units in the front bedroom on a 4BR unit, the outdoor unit may be installed in the front such that the line set will run along the front wall. It is not permitted to run a line set or any other connection above the roof nor for it to penetrate the roof.

For aesthetic reasons, the exterior routing of the line sets will be carefully reviewed before an application will be approved.

Aesthetical considerations

The following aesthetical considerations for the conduit shall apply only to exterior-visible areas of the unit.

For front and back walls, the exterior-visible portion is defined as part of the wall above the fence line (typically 7 feet above ground).

For side and carport walls, the entire wall is considered exterior-visible.

Non-exterior visible areas include the atrium walls (on 3BR units) and all interior walls.

- The installation of exterior-visible conduit shall be done via rectangular conduit shaped similarly to the existing rain downspouts attached to the exterior of the buildings in our community, or, a circular pipe or tubing of approximately ¾" diameter identical to existing water pipes. If multiple ¾" pipes are required, they shall be encased in a rectangular conduit.
- Exterior conduit in a carport or side wall shall be run near the ground (similar to where exterior water pipes have been routed).

- Exterior-visible conduit that runs vertically along a wall shall run all the way up to the roof overhang so it blends in and looks like a rain downspout. It should not terminate and penetrate the middle section of the wall, only near the edge of the wall near the roof overhang. (To be clear, the conduit will not extend into the overhang or penetrate the roof. It is just intended to be installed so it looks like a rain downspout while concealing the inner piping.)
- Exterior-visible conduit shall run vertically as far as possible. If required, horizontal elements shall be run below the fence line or run along the roof overhang boundary, but not along the exterior-visible surface of a wall. It is preferable that the conduit cover aligns with existing vertical lines on the exterior walls.
- If two conduits are required, they shall either be positioned at least 10 feet apart from each other, or some other solution be devised to prevent multiple vertical elements less than 10 feet apart along the facade of the building (e.g., a larger cover that is rectangular, goes from floor to ceiling, and matches the same wood-lined texture as the existing siding).
- All exterior-visible components, conduits or covers shall be painted to match the color of the exterior wall (using a HOA-approved exterior paint).

Structural considerations

The installation of all components (indoor unit, outdoor unit, and conduit) shall not compromise the structural integrity of the walls nor the roof. While conduits and the mounting of the indoor unit, may require wall penetration and be connected to overhead beams, any conduit shall not penetrate the roof nor compromise the structural integrity of beams and joists supporting the walls and roof.

Acoustics

For the consideration of neighboring units, the acoustic rating, sound pressure level or noise level of the outdoor unit, in the cooling mode, should be as low as possible given current offerings in the market. The maximum allowed noise level per the manufacturer's specification shall be 55 dB (or 55 dBA) in the cooling mode.

Electrical

While smaller mini-split A/C units may run off a 120V circuit, most larger units will require a 240V circuit.

A dedicated electrical circuit and circuit breaker should be assigned or installed for the outdoor unit.

The total load on the sub-panel after the addition of the A/C unit shall not exceed the 90A maximum for the unit, which is set by the wiring limit to each unit.

All electrical wiring additions, modifications and connections to power the A/C unit shall be to code and performed by a qualified and licensed electrician.

Any exterior-visible conduits used for electrical wiring shall follow the aesthetic and structural considerations listed above.

Permit

All work, including A/C and electrical, shall comply with Santa Clara city code and requirements. The necessary permits shall be applied for by the homeowner or the contractor.

Application Process and Timeline

An application for installing a mini-split A/C is required. It should contain:

- An itemized description of all the outdoor and indoor units (including manufacturer and model numbers) and an online web link to the respective specification sheets. The specification sheet should state the noise level and required circuit breaker load capacity for the outdoor unit. This is to verify compliance to the 55 dB noise level limit.
- The name, contact information and license # of the installation contractor(s).
- The certificate of liability insurance certificate of the installation contractor(s).
- One or more detailed diagrams showing the position and size of all the outdoor unit(s), conduits (pipes, covers, line sets), electrical connection boxes, and any other exterior visible features. It is preferable that a clear photo of the location (i.e. of the wall) be shown, then the position and size (to scale) of the relevant components to be added, be drawn or rendered on top of the photo. Please use a color legend or label to identify the components. Please add a marking line denoting the height of the fence line to the diagram as well. This diagram shall accurately reflect the planned position, size, and routing and be agreed to be adhered to by the contractor as well before submission, rather than being an approximation.
- Photos of the relevant exterior-visible components, i.e. line set cover, electrical box, pipes, etc. (can be taken from another installation or from the spec sheet of the component).
- An electrical load table, showing after any A/C-required modifications, for each circuit breaker in the sub-panel, its amperage. Please call out the circuit breaker dedicated to supplying the A/C unit. This is to verify compliance to the 90A limit for the external main electrical wiring to the unit.

Following the approval of the application, please update the HOA with the date/time of the installation, so that at their discretion, officers of the HOA or their assigned members can view the installation to its completion.

The HOA would prefer to avoid any post-installation concerns regarding any aspects of an A/C installation. An inspection for adherence of exterior-visible installations to the prior-submitted plan may be conducted. The homeowner will be notified of any concerns regarding the installation no later than 30 days after the date of installation.

Maintenance

- Homeowner must maintain condition of the A/C unit and related components (includes appearance, noise level, safety)
- The HOA may request homeowner action when condition of A/C unit and related components are unacceptable
- The HOA may authorize removal of A/C units and related components if conditions warrant

- Installations are permanent (a removal or modification to externally-visible components will require a new exterior modifications request application)
- Homeowner is responsible for any damage to personal and/or community property caused by the installation or operation of their A/C unit(s)